

What is in the Waterbug Testing Kit?



Dissecting
Microscope



Bulb Pipettes



6-dish plate



Net



Gumboots



Spoons and
Ladle



White Tubs



Bucket

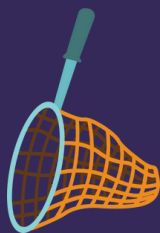
How to Collect your own waterbug samples from a waterway

1 Entering a Waterway (always with supervision from an adult)

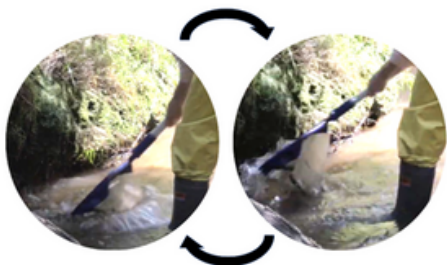
- DO NOT enter if the water is too deep.
- DO NOT enter if water is moving fast.
- Find the safest spot to enter the water.
- Use your net to check the depth and look for any hidden sinkholes.
- Stay in shallow areas



2 Collect Samples



Use the net to collect water in the water trays with one of these methods



To collect bugs near the edges or plants, place your net in the water and scoop it up



To collect bugs near rocks or in open areas, hold your net in front of you in the water and slowly walk backwards

How to use the Waterbug Testing Kit to conduct an investigation

1 Separate the Waterbugs

- Use spoons to collect waterbugs into a six-dish tray. Separate individual bugs into their own section.



2 Identify the Waterbugs

1. Look at the Waterbugs under the dissecting microscope



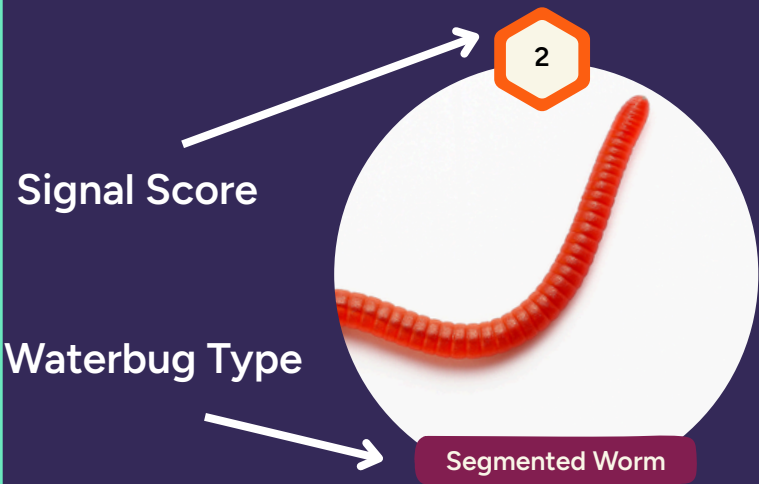
2. Use GTAC's Waterbug ID Resource to identify each waterbug



<https://tinyurl.com/mnryx62u>

3 Record the Signal Score of each waterbug

For each waterbug you find, record what type of bug it is, how many times you see each bug (tally) and its signal score (number from 1 to 10). Use the table provided by GTAC.



Keep count how many times you see the same type of bug

Type of Waterbug	Tally	Signal Score
Segmented Worm	III	2

4 Calculate totals

Calculate the total types of waterbugs you saw (blue box) and the total signal scores (green box)

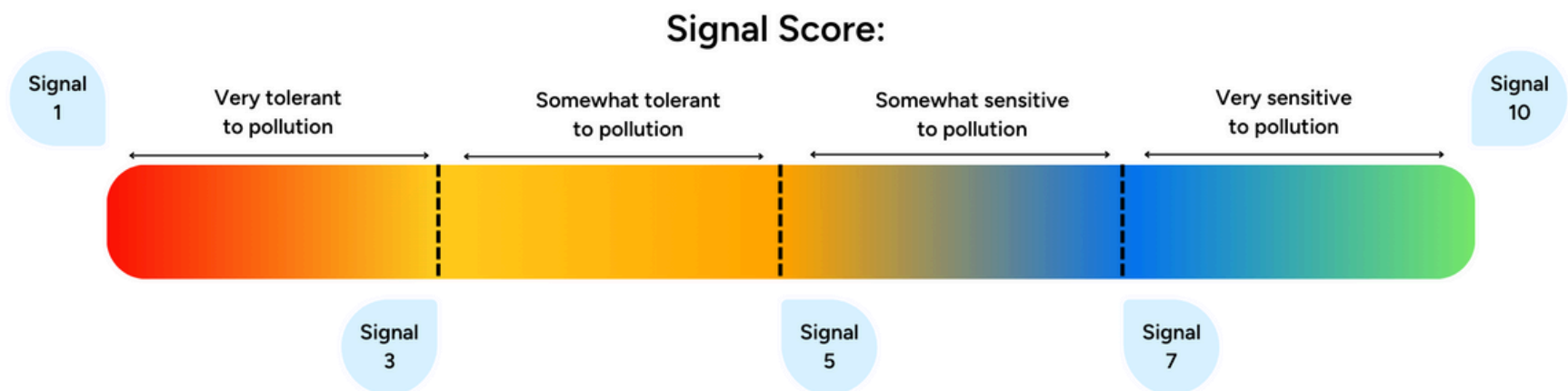
Type of Waterbug	Tally	Signal Score	Tally x Signal Score
Dragonfly Nymph	I = 1	3	3 x 1 = 3
Water Strider	II = 2	4	4 x 2 = 8
	Total Tally: 3		Total Signal Score: 3 + 8 = 11

How to analyse the data and reflect on the results

1

Understanding the Signal Score

Waterbugs are given a signal score rating which can give us information about the quality of the water they live in.



- **Sensitive waterbugs** are like picky house guests. They only live in clean, healthy water. If the water gets polluted or dirty, they can't survive.
- **Tolerant waterbugs** are tougher. They don't mind if the water is a bit dirty or polluted. They can live in places where sensitive bugs cannot.

Note: Tolerant waterbugs can live in all sorts of water, so may be present in both unpolluted and polluted water. Sensitive bugs, however, can only live in clean, unpolluted water.

2

Healthy Waterways

Waterways with an average signal score of 6 or more fall within healthy levels of pollution. A score between 8-10 means a high level of bugs that are sensitive to pollution therefore the waterway is not polluted.

< 6

Polluted

6 ↔ 8

Minimal Pollution

> 8

Not Polluted

3

Use your results to calculate the Average Signal Score:



- Add the totals on your data table (in the blue and green boxes)

Type of Waterbug Find out it's name on the App or ID chart	Tally How many?	Signal Score Find out it's Signal Score on the Waterbugs App	Tally x Signal Score
Dragonfly Nymph	I = 1	3	$1 \times 3 = 3$
Water Strider	II = 2	4	$2 \times 4 = 8$
Freshwater Prawn	IIII = 4	4	$4 \times 4 = 16$
	Total Tally: 7		Total Signal Score: $3 + 8 + 16 = 27$

- Calculate the Average Signal Score using the equation (copy from the blue and green boxes)

$$\text{Average Score} = \frac{\text{Total Signal Score}}{\text{Total Tally (Total Number of Waterbugs)}}$$

$$\text{Average Score} = \frac{27}{7}$$

$$= 3.86$$

4 What does your result tell you? Think back to your original question (from the 'Ask' Stage)

An average score of 3.86 indicates that most of the bugs in this waterway are tolerant to pollution. Because of this, the water is likely to be quite polluted.

This result is based on a very small number of waterbugs and so we cannot say for sure if this result is correct. It would be best to collect more samples from different parts of the waterway to get a more realistic result.